

Module 13: How Populations Evolve — Questions

Introduction

The following questions are designed to test your understanding of microevolution, the mechanisms that drive changes in allele frequencies, and the conceptual framework of the Hardy-Weinberg equilibrium. Use this as a study tool to ensure you have grasped the core forces that shape populations over time.

Practice Questions

1. What is the official definition of microevolution in biology?
2. What are the five main mechanisms (or "fingers") of microevolution?
3. How does mutation act as the ultimate source of all genetic variation in a population?
4. What is the core difference between gene flow and genetic drift?
5. Why does genetic drift have a much more significant and unpredictable impact on smaller populations compared to larger ones?
6. Describe the Bottleneck Effect and provide a real-world or hypothetical example of how it reduces genetic diversity.
7. How does the Founder Effect lead to a new population with limited genetic diversity, and how does this differ from a bottleneck?
8. Which mechanism of microevolution is the *only* one that consistently leads to adaptation, and why?
9. What is directional selection, and what kind of environmental changes typically trigger it?
10. How does stabilizing selection affect intermediate versus extreme variations of a physical trait?
11. Describe a scenario that represents disruptive selection. Why are individuals with intermediate traits at a disadvantage in this scenario?
12. What is non-random mating (sexual selection) and how can it change allele frequencies over time?
13. Define the Hardy-Weinberg equilibrium conceptually. What does it represent?

14. Why is a population in perfect Hardy-Weinberg equilibrium rarely (if ever) found in nature?
15. How do scientists use the Hardy-Weinberg equilibrium as a baseline to measure how fast a population is evolving?